4. (b) Advanced Data Modelling and Form Design

Aim

To extend the lost and found application with advanced field types and form features: a lookup field, a subform, an image link field, a radio field and a default value.

Theory

Advanced data modelling links forms to one another and groups repeating data. A lookup field stores a reference to a record of another form, so a claim points to an existing found item instead of copying its details. A subform keeps repeating child rows (here the actions taken on a claim) inside the parent form. A URL or image field holds a link to, or a photograph of, the item.

Feature	Field / Form	Purpose in the application
Lookup	Claim_Requests → Found Item	A claim refers to one found item; one found item can have several claims
Subform	Claim_Requests → SubForm (Action Date, Action Taken, Remarks)	Keeps a dated trail of the actions taken on each claim
Image link field	Item Image on the Lost and Found forms	Link to a photograph of the item
Radio field	Found_Items → Status (Available, Claimed, Other)	Fixed status values
Default value	Subform Action Date = current date	First action row is created with today's date

Procedure

1. Open the Claim_Requests form in the Form Builder and add a Lookup field named Found Item that takes its values from the Found_Items form.

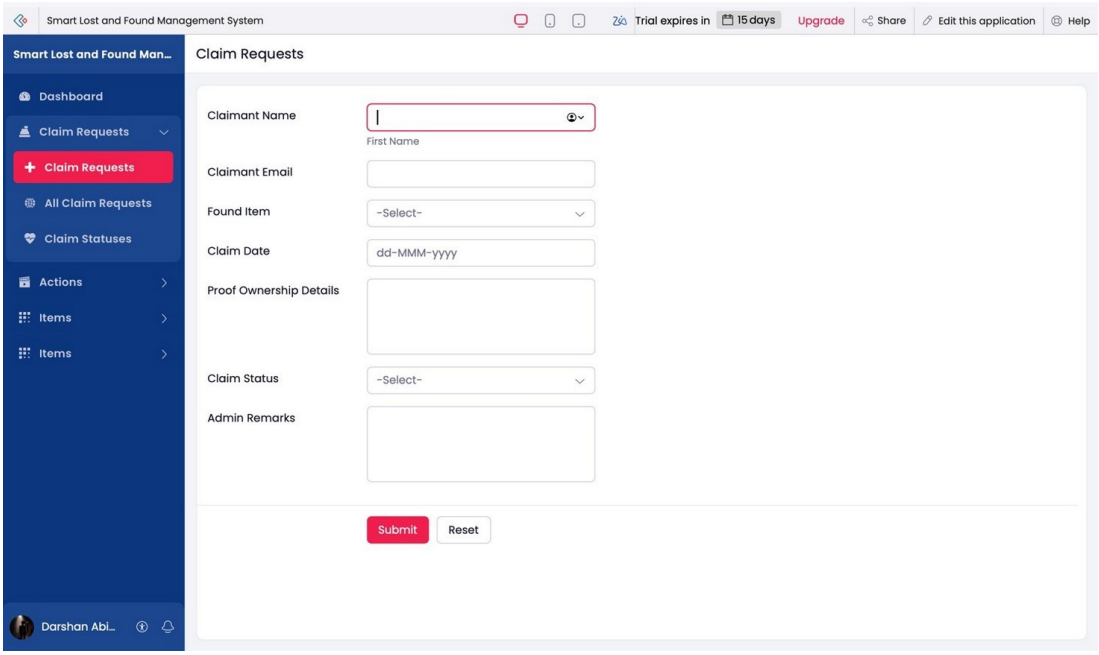


Fig. 4.1: Lookup field Found Item on the Claim_Requests form

2. Add a Subform with the fields Action Date (default: current date), Action Taken (drop down, for example Claim Submitted) and Remarks (multi line).
3. Add the image link field Item Image and the radio field Status on the found item form.

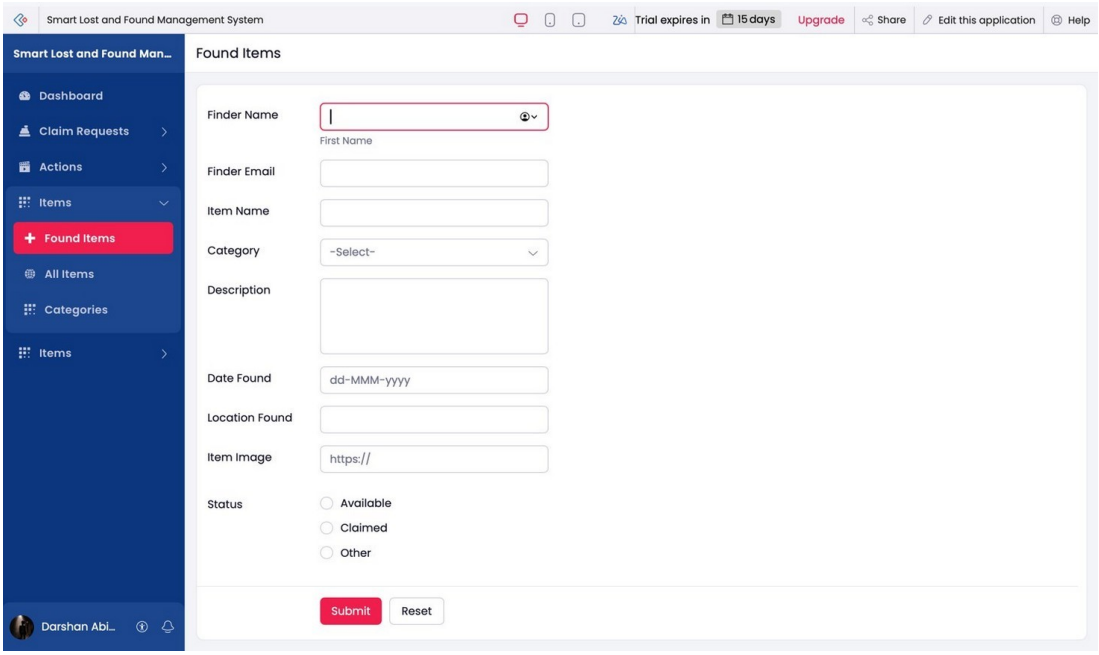


Fig. 4.2: Found_Items form with the Item Image link field and the Status radio field

4. Save the form, open it in live mode and select a found item in the lookup. The subform shows the first action row with the current date.

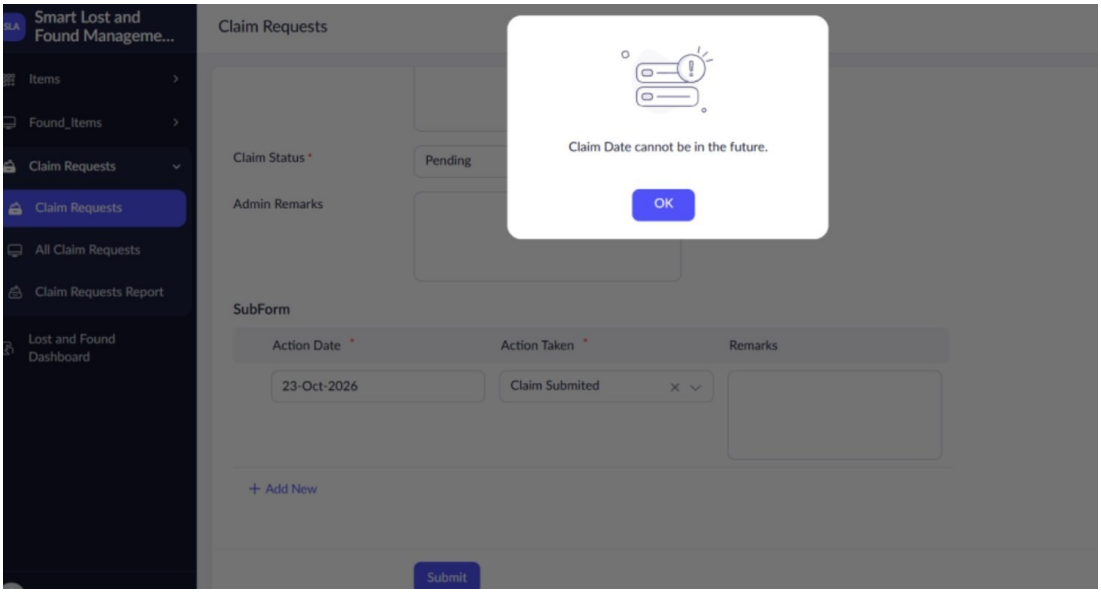


Fig. 4.3: Claim_Requests form in live mode showing the subform row

5. Submit a claim and open the Claim Requests Report to check that the claim shows the linked found item.

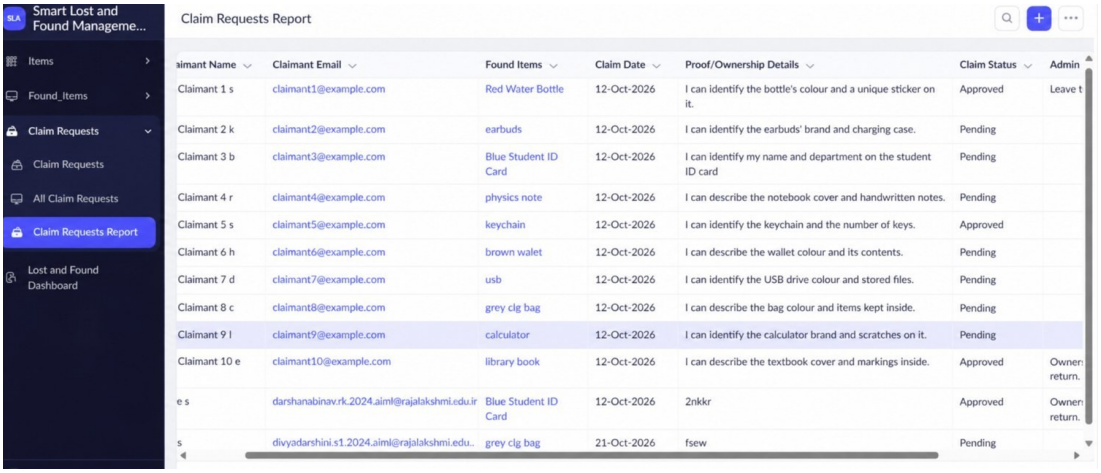


Fig. 4.4: Claim Requests Report with the Found Items lookup column

Result

The lookup field and the subform were implemented in the Claim_Requests form. A claim now refers to an existing found item and carries its own list of dated actions, and the first action row takes the current date by default. The Found_Items form uses an image link field for the item photograph and a radio field for its status.